

All user stories are based on open interviews (especially with supervisors) + line visit where we had the opportunity to see the line and ask further stakeholders. For example, it was told that recall is the most important metric (further explained in the report)

User Story 1: Quality Prediction	
<i>As a quality engineer, I want to automatically predict product quality during SMT production so that I can detect defective products early and reduce rework and waste.</i>	
Acceptance criteria	- trained ML model predicts product quality (with classes no error, suspect or error) with high evaluation metrics (e.g. > Recall class 2 (error) = 1, before final inspections (AOI, EOL tests) - User can view the top X influencing features for each prediction (Explainable AI) - Results can be exported for a report
ML Task	- Supervised learning (i.e. XGBoost) - Feature engineering - XAI

User Story 2: Feature-Explanations for predictions	
<i>As a line operator, I want to see the top features influencing a prediction that involves the class “error” so that I can quickly understand possible causes and adjust the process.</i>	
Acceptance criteria	- SHAP provides feature importances per prediction - Display in the UI with names of the features - Results can be exported for a report
ML Task	Explainable AI (e.g. SHAP)

User Story 3: What-if Analysis
<i>As a data scientist, I want to simulate the impact of changes to individual input features on the model prediction (e.g. through SHAP-based what-if analyses)</i>

so that I can test process hypotheses and identify potential levers for quality improvement.

Acceptance criteria

- What-if scenarios can be performed on stored test data
- Changes in predictions are visualised (permutational feature importance)

ML Task

- Explainable AI (i.e. permutational feature importance)
- Results can be exported for a report

User Story 4: Data Quality

As a data engineer, I want to automatically check whether data is valid before training (e.g. no columns with only zero values) so that I can ensure that the model is not trained on incorrect data.

Acceptance criteria

- Plausibility checks are performed during preprocessing
- Notable features or rows can be extracted

ML Task

- (automatic) feature checks
- Data validation
- Techniques like balancing, encoding etc. taken into account